M-SAP-WP-1.2 Client-Server communication with SAP ABAP

Level	Duration	Workload
Master	1 semester	**Total:** 150 h, thereof
Module supervisor	**Frequency**	• 28 h physical synchronous contact hours
see Curriculum	Twice a year	• 122 h independent study hours
Language of instruction	**Weekly contact hours**	
English	2	
ECTS points		
6		

Learning Outcomes

After successful completion of the module, students are able to:

- Gain a comprehensive understanding of the ABAP (Advanced Business Application Programming) language, including its syntax, data types, and basic programming structures.
- Learn to develop, debug, and maintain ABAP programs using the SAP development environment, ensuring efficiency and adherence to best practices.
- Understand how ABAP integrates with various SAP modules and systems, enabling the development of custom applications that enhance SAP functionality.
- Acquire skills in designing user-friendly interfaces using ABAP Dynpro (dynamic programming) and SAP Web Dynpro for both traditional and web-based applications.
- Delve into advanced ABAP programming techniques such as modularization, performance tuning, and exception handling to develop robust and scalable applications.
- Apply theoretical knowledge through hands-on exercises and real-world scenarios to solve practical business problems using ABAP programming.

Module Contents

1. **Introduction to ABAP Programming**
   - Overview of ABAP language and its role in SAP
   - Basic syntax and structure of ABAP programs
   - Understanding ABAP development tools (e.g., ABAP Workbench, Eclipse)

2. **Data Types and Data Objects**
   - Fundamental data types and objects in ABAP
   - Internal tables, structures, and arrays
   - Type conversion and casting

3. **Modularization Techniques**
   - Creation and use of subroutines and function modules
   - Understanding and implementing macros
   - Benefits and use of classes and methods (Object-Oriented ABAP)

4. **ABAP Data Dictionary**
   - Purpose and functionality of the Data Dictionary
   - Defining and managing tables, views, indexes, and data elements
   - Utilizing search helps and lock objects

5. **User Interface Programming**
   - Basics of designing user interfaces with SAP Dynpro

- Introduction to SAP Web Dynpro for web-based applications
- Event handling and screen flow logic

6. **Enhancements and Modifications**
   - Concepts of SAP enhancements and modifications
   - Using user exits and customer exits for custom development
   - Implementing Business Add-Ins (BADIs) and enhancement points

7. **Performance Tuning and Best Practices**
   - Techniques for optimizing ABAP code performance
   - Best practices for efficient and maintainable ABAP programming
   - Analyzing and improving program performance with SAP tools (e.g., SQL Trace, Runtime Analysis)

Learning activities and teaching methods

Physical synchronous contact hours:
Lecture, supported by exercises and case studies with discussion.

Independent study hours:
Self-directed preparation and follow-up of online seminars, literature study, exam preparation

SAP systems:

For this lecture, SAP ABAP will be used.

Participation requirements

No participation requirements are recommended.

Assessment methods and criteria

Project work
The students complete this module with a project report (i.e., choose one aspect of the programming language and write a report of 10 to 15 pages).

A passing grade in this module is achieved when the overall grade is greater than or equal to 4.0.

Connection to other programs and modules

This module is taught exclusively in this degree program.

Share of module grade for final grade

6/120 Credit Points

Literature

Mandatory Reading

- Horst Keller, "ABAP Objects: Introduction to Programming SAP Applications," 4th Edition, SAP PRESS, 2020.

Further Reading

- Paul Hardy, "ABAP to the Future," 3rd Edition, SAP PRESS, 2019.
- Thomas Schneider, "Getting Started with ABAP: Beginner's Guide to SAP ABAP," SAP PRESS, 2016.

- James Wood, "Object-Oriented Programming with ABAP Objects," 2nd Edition, SAP PRESS, 2015.
- Bernd Matzke, "SAP ABAP: Hands-on Test Projects with Business Scenarios," Galileo Press, 2016.